Bocconi

CORPORATE FINANCE 30017

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Applying the CAPM (Basics)

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Agenda

ESTIMATING DISCOUNT FACTORS

METHODOLOGY

IMPLEMENTATION

CAPM AND COST OF EQUITY

ESTIMATING DISCOUNT FACTORS

Mother of all Questions in Asset Pricing

What is the fair price for asset $\boldsymbol{i}$? $\Leftrightarrow$ At what rate to discount cash flows from asset $\boldsymbol{i}$?

$$P_{i,0} = \sum_{t=1}^{\infty} \frac{\boldsymbol{E}[c_{i,t}]}{\prod_{h=1}^{t}(1+\delta_{i,t})}$$

Note for $\delta_{i,t} = \delta_i \ \forall t$:

$$P_{i,0} = \sum_{t=1}^{\infty} \frac{\boldsymbol{E}[c_{i,t}]}{(1+\delta)^t}$$

One Model to Price them All

— Asset's discount rate is equivalent to expected return:

$$\delta_{i,1} = \frac{\boldsymbol{E}[d_{i,1}]}{P_{i,0}} + \frac{\boldsymbol{E}[P_{i,1}] - P_{i,0}}{P_{i,0}} \equiv \boldsymbol{E}[r_{i,1}]$$

— …for which the CAPM makes the following prediction:

$$\boldsymbol{E}[r_{i,1}] = r_{f,1} + \beta_i\left(\boldsymbol{E}[r_{m,1}] - r_{f,1}\right)$$

- If only we knew β_i …

METHODOLOGY

Excursus: One layer deeper in the CAPM

Not exam relevant

— Under CAPM assumptions, optimality of the market portfolio implies for each asset i:

$$\boldsymbol{E}[r_{i,t}] - r_{f,t} = \lambda Cov(r_{i,t} - r_{f,t}, r_{m,t} - r_{f,t})$$

- For market portfolio itself: $\boldsymbol{E}[r_{m,t}] - r_{f,t} = \lambda Var(r_{m,t} - r_{f,t}) \Leftrightarrow \lambda = \frac{\boldsymbol{E}[r_{m,t}] - r_{f,t}}{Var(r_{m,t} - r_{f,t})}$

— Substitute λ:

$$\boldsymbol{E}[r_{i,t}] - r_{f,t} = \boxed{\frac{Cov(r_{i,t} - r_{f,t}, r_{m,t} - r_{f,t})}{Var(r_{m,t} - r_{f,t})}} (\boldsymbol{E}[r_{m,t}] - r_{f,t})$$

"β_i"

Estimate β_i

— Define excess returns (to ease notation):

- $x_t = r_{i,t} - r_{f,t}$
- $y_t = r_{m,t} - r_{f,t}$

— Manually:

$$\widehat{\beta}_i = \frac{\sum_{t=0}^{T}(x_t - T^{-1}\sum_t x_t)(y_t - T^{-1}\sum_t y_t)}{\sum_{t=0}^{T}(y_t - T^{-1}\sum_t y_t)^2} \xrightarrow{\mathrm{T}\to\infty} \frac{Cov(r_{i,t} - r_{f,t}, r_{m,t} - r_{f,t})}{Var(r_{m,t} - r_{f,t})}$$

— Via univariate regression (mathematically equivalent): $x_t = \alpha + \beta_i y_t + e_t$

$$\widehat{\beta}_i \xrightarrow{\mathrm{T}\to\infty} \beta_i = \frac{Cov(r_{i,t} - r_{f,t}, r_{m,t} - r_{f,t})}{Var(r_{m,t} - r_{f,t})}$$

IMPLEMENTATION

Empirical Choices

- Pick an asset i!

- Which frequency Δt (daily, monthly, quarterly, annually,...)?
- Which sample size T?

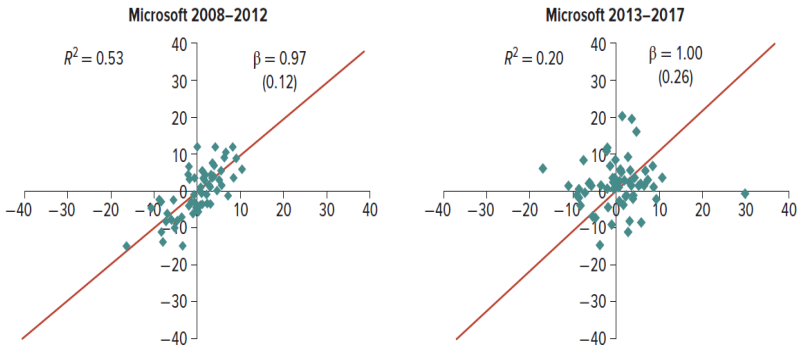
- How to measure the risk free return $r_{f,t}$?
- How to measure the market return $r_{m,t}$?

Some basic practical tips

- Do not regress prices on prices (always regress returns on returns)
- Returns should be based on prices adjusted for stock splits and dividends
- Proxies of the market portfolio should be value-weighted stock market indices, such as the CRSP value-weighted Market Return, or the SP500
- You need a lot of data (i.e. long time series) to reliably estimate beta:
  - One way to achieve this is to use a low data frequency (e.g. monthly returns) and go back far in time (e.g. 5 years)—the downside here is that the true relationship between the stock and the market may have changed
  - Another way to achieve this is to use a high data frequency (e.g. daily returns) and consider only the recent past (e.g. 1yr)—the downside here is that higher frequency data may be subject to illiquidity bias, or capture non-representative time periods

Running Regressions in Excel

- You are more than welcome to use other software
- But excel is intuitive
  - You already know
    - VAR.P and VAR.S
    - STDEV.P and STDEV.S
    - COVAR and COVARIANCE.P
    - CORREL
    - AVERAGE
  - Now we will use
    - RSQ
    - SLOPE



What are NVIDIA's vs. Apples Expected Returns?

1. Estimate equity betas
   - Download stock return data from WRDS
     - CRSP, Stock/Security Files/Monthly Stock File
     - Select firms, suitable time window
     - Select: ticker, permco, ret
   - Download risk-free and market return data from WRDS
     - Fama-French Portfolios, 3 Factors Plus Momentum Monthly Frequency
     - Set same data range
     - Select: mktrf, rf
   - Import to Excel, merge manually
   - Compute β via SLOPE (compare to ratio of COVAR and VAR.S)
     - Be fancy: Make scatter plot of excess returns;: "Add Chart Elements": "Trend Line, Linear"
2. Compute expected returns under current risk free rates and assumed 5% market risk premium

CAPM AND COST OF EQUITY

Using CAPM for Stocks

- CAPM generally applies to **all (publicly traded) assets**
- Very important use case in practice: **Stocks**
  - Investor's perspective: *What is the fair value of a given stock?*
  - CFO's perspective: *What is the company's **cost of equity** imposed by the stock market?*

Measuring Cost of Equity

- CFO: *What discount rate to use for PVs of (equity-financed) investment opportunities?*
- Cost of Equity is discount rate equating PV of expected dividends with stock's market price
  - I.e., IRR of a stock investment

$$P_{i,0} = \sum_{t=1}^{\infty} \frac{\boldsymbol{E}[d_{i,t}]}{\prod_{h=1}^{t}(1+\delta_{i,t})}$$

- Equivalent to expected return of that stock:

$$\boldsymbol{\delta_{i,1}} = \frac{\boldsymbol{E}[d_{i,1}]}{P_{i,0}} + \frac{\boldsymbol{E}[P_{i,1}] - P_{i,0}}{P_{i,0}} \equiv \boldsymbol{E}[r_{i,1}]$$

- Hence, two options to estimate cost of equity:
  - CFO predicts all future dividends (hoping to match market expectations), takes price, solves for IRR
  - CFO applies CAPM (trusting assumptions hold and β_i is estimated precisely)